

DATA 260 Homework 2 Report

Student and Configuration

- Student: Vrishin Dharmesh Kunnatham Parambath
- SID4: 8844
- PORT_BASE: 8744
- PREFIX: s8844
- SEED: 8844
- VERIFY_SEED: 268844
- DOMAIN_ID: 4
- Assigned domain: Open-source package vulnerabilities
- Model: qwen3:8b
- Hardware: Windows 11 laptop
- Final tag: hw2
- GitHub repository: <https://github.com/Viki2183/data260-8844>

Part 1 — HTML and CSS

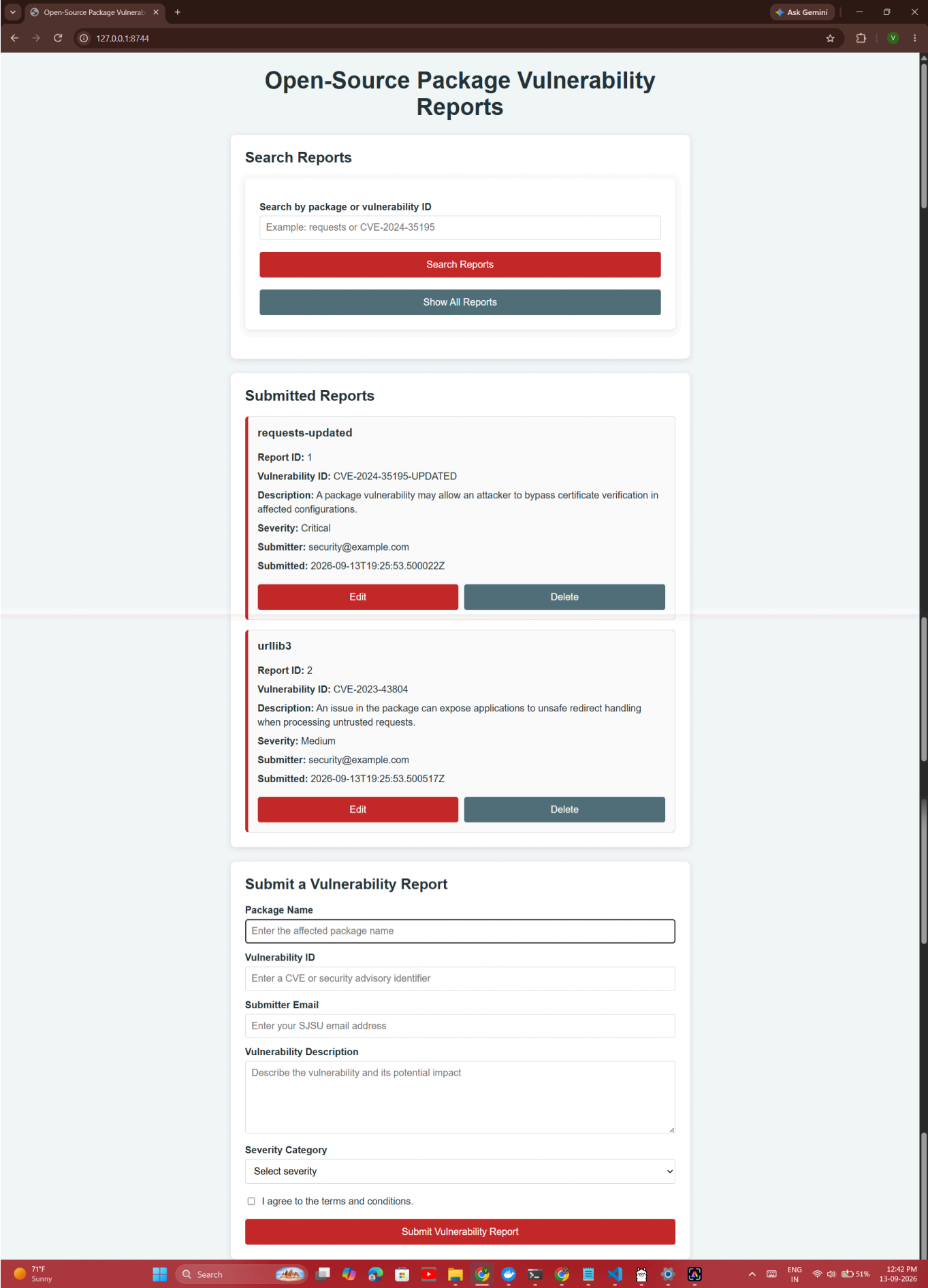
The HW1 vulnerability form was extended with a responsive layout. The page uses an external `styles.css` file and includes a mobile media query for narrow screens, including a 375px-wide display.

The interface includes:

- A vulnerability report submission form
- A search form
- Report cards
- Edit and delete buttons
- A visible loading state
- A visible empty state
- A visible error state

The page was tested through the Dockerized application at:

<http://127.0.0.1:8744>



Open-Source Package Vulnerability Reports

Search Reports

Search by package or vulnerability ID

Example: requests or CVE-2024-35195

Search Reports

Show All Reports

Submitted Reports

requests-updated

Report ID: 1

Vulnerability ID: CVE-2024-35195-UPDATED

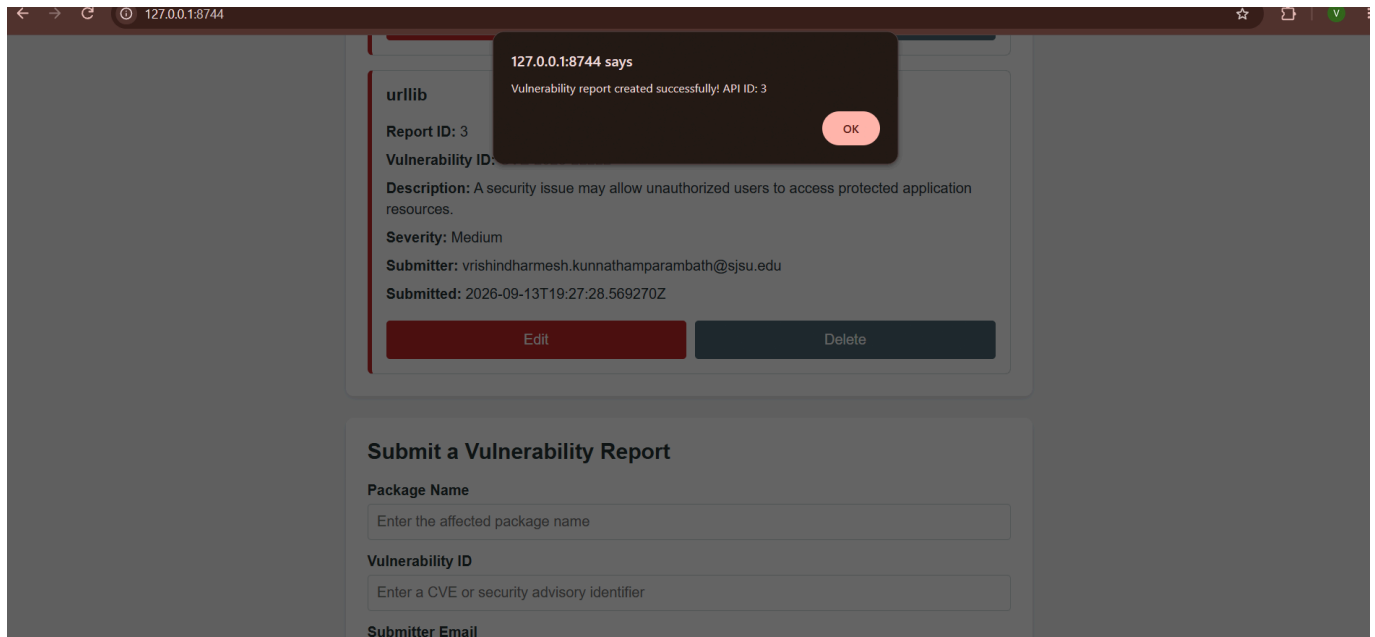
Description: A package vulnerability may allow an attacker to bypass certificate verification in affected

Part 2 — FastAPI

The FastAPI backend runs on `PORT_BASE` 8744 and provides REST endpoints for vulnerability reports.

Part 2.1 — Create a Record

The frontend sends a POST request to `/api/vulnerability-reports`. A new report was created successfully and returned API ID 3.



Part 2.2 — Update Record ID 1

The report with ID 1 was edited through the webpage. Its package name was changed to `requests-updated`, its vulnerability ID was changed to `CVE-2024-35195-UPDATED`, and its severity was changed to `Critical`.

Edit Report 1

Package Name

requests-updated

Vulnerability ID

CVE-2024-35195-UPDATED

Submitter Email

security@example.com

Vulnerability Description

A package vulnerability may allow an attacker to bypass certificate verification in affected configurations.

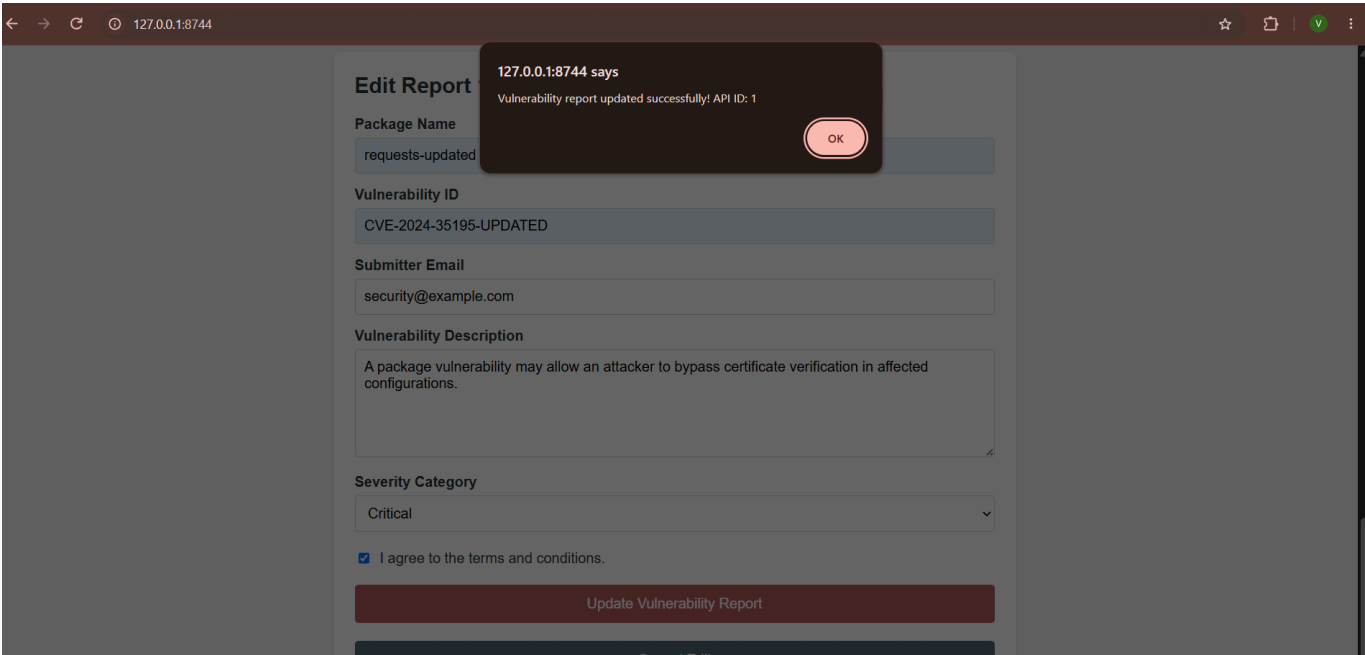
Severity Category

Critical

☒ I agree to the terms and conditions.

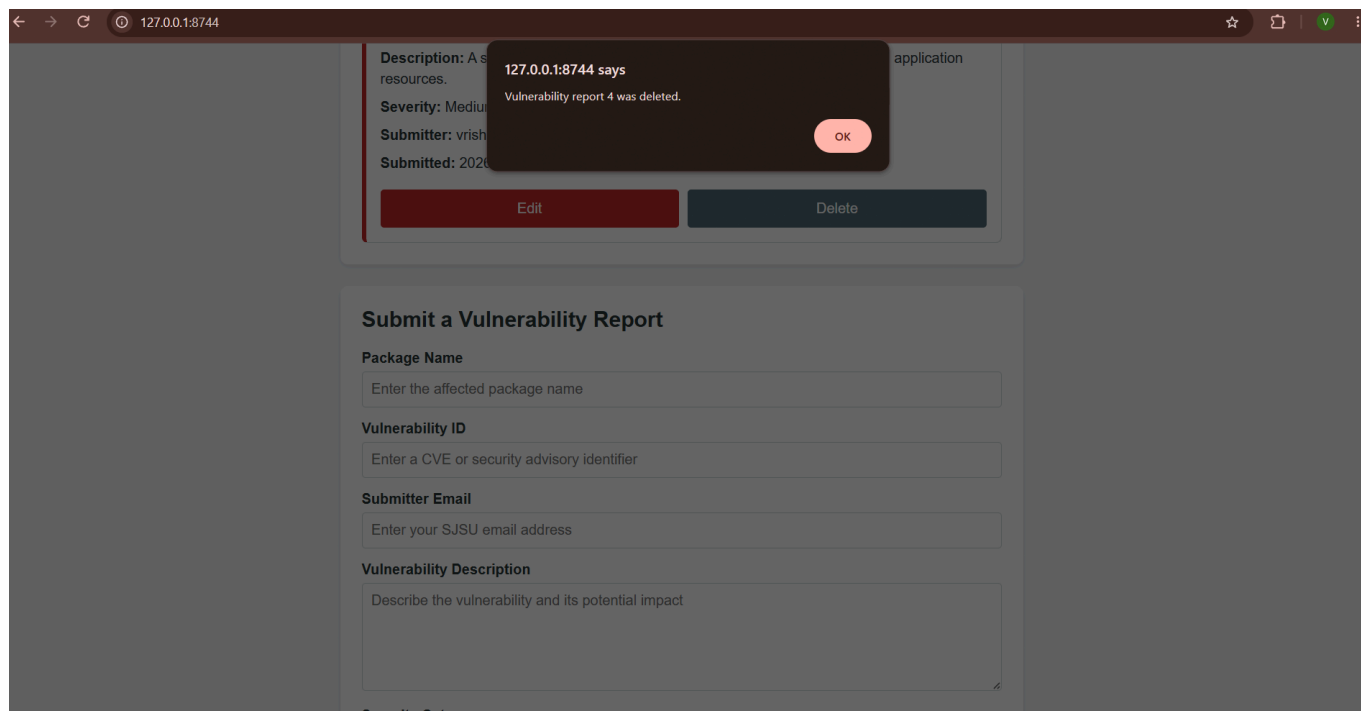
Update Vulnerability Report

Cancel Edit



Part 2.3 — Delete the Highest-ID Record

The temporary highest-ID report, Report ID 4, was deleted through the webpage. The success message confirmed the deletion, and the refreshed report list showed that Report ID 4 was no longer present.



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Search Reports

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Search Reports

Show All Reports

Submitted Reports

requests-updated

Report ID: 1

Vulnerability ID: CVE-2024-35195-UPDATED

Description: A package vulnerability may allow an attacker to bypass certificate verification in affected configurations.

Severity: Critical

Submitter: security@example.com

Submitted: 2026-09-13T19:25:53.500022Z

EditDelete

urllib3

Report ID: 2

Vulnerability ID: CVE-2023-43804

Description: An issue in the package can expose applications to unsafe redirect handling when processing untrusted requests.

Severity: Medium

Submitter: security@example.com

Submitted: 2026-09-13T19:25:53.500517Z

EditDelete

Submit a Vulnerability Report

Package Name

Enter the affected package name

Vulnerability ID

Enter a CVE or security advisory identifier

Submitter Email

Enter your SJSU email address

Vulnerability Description

Describe the vulnerability and its potential impact

Severity Category

Select severity

☐ I agree to the terms and conditions.

Submit Vulnerability Report

72°F Sunny

Search

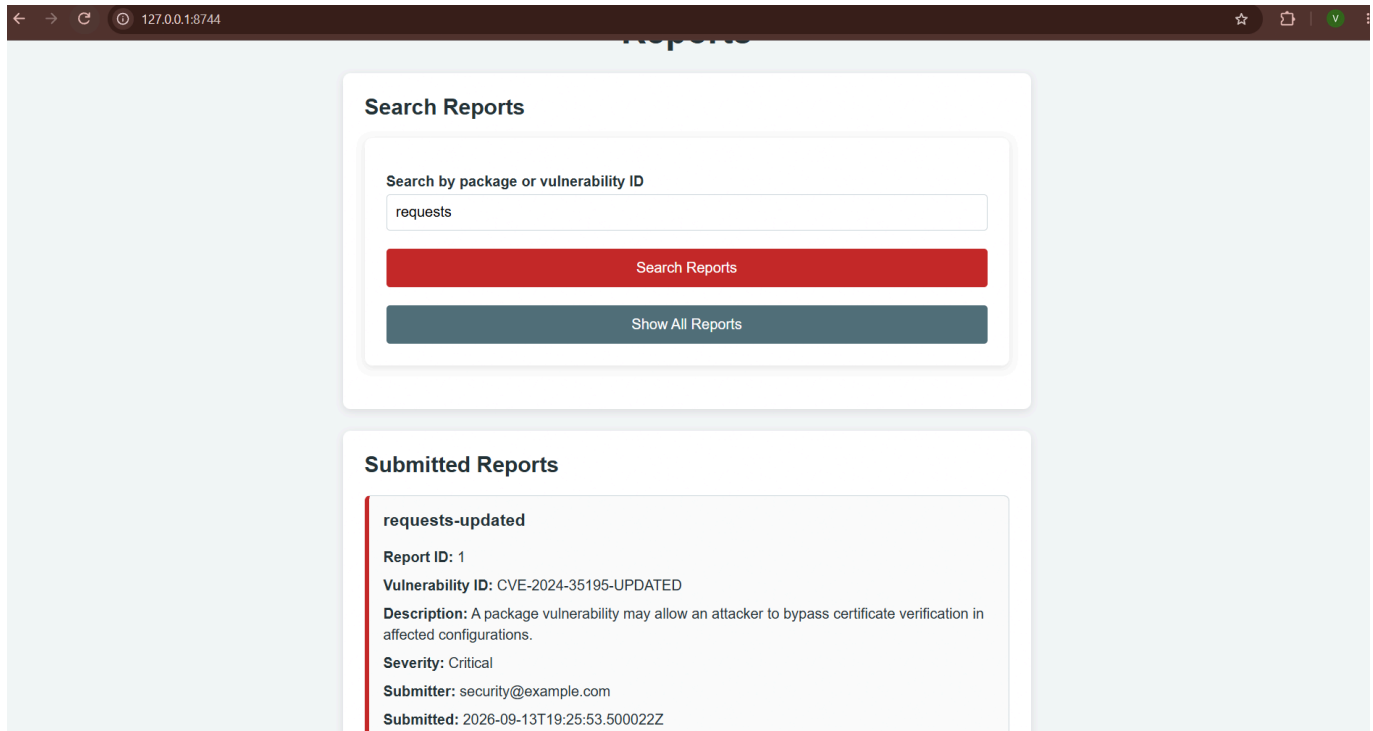
ENG IN

60%

04:06 PM 13-09-2026

Part 2.4 — Search

Searching for **requests** returned only the matching report.



Part 3 — Stateful Agent Graph

The sequential HW1 workflow was extended into a stateful LangGraph workflow in [code/hw2_graph.py](#).

The graph contains:

- [AgentState](#), which stores shared workflow memory
- A Planner node
- A Reviewer node
- A Supervisor node
- Conditional routing logic
- A turn counter to prevent infinite loops
- A retry path when validation or review fails

The Planner output is validated with Pydantic. The schema requires exactly three distinct string tags, each between 3 and 30 characters, and a summary containing no more than 25 words.

The graph was tested with the local [qwen3:8b](#) model. The test completed in 3 supervisor turns, and the Reviewer approved the output.

The final output contained exactly three tags:

- [AuthenticationBypass](#)
- [RequestsPackage](#)
- [UnauthorizedAccess](#)

The generated summary was within the 25-word limit.

```

C:\Projects\data260-SB44\venv\Lib\site-packages\langgraph\cache\base\_init_.py:8: LangChainPendingDeprecationWarning: The default value of 'allowed_objects' will change in a future version. Pass an explicit value (e.g., allowed_objects='messages' or allowed_objects='core') to suppress this warning.
  from langgraph.checkpoint.serde.jsonplus import JsonPlusSerializer
-- NODE: Supervisor (turn 1/10) ---
-- NODE: Planner ---
turn token usage: input=160, output=258, total=418
Lanner produced valid output in 76572 ms.
-- NODE: Supervisor (turn 2/10) ---
-- NODE: Reviewer ---
turn token usage: input=160, output=344, total=513

{"approved": true,
 "issues": [],
 "latency_ms": 82083.95}
-- NODE: Supervisor (turn 3/10) ---
-- GRAPH RESULT ---
{"title": "Critical Authentication Bypass in Requests Package",
 "email": "vishindharmesh.kunnathparamath@rsu.edu",
 "model": "gpt4o",
 "temperature": 0.8,
 "max_turns": 10,
 "turn_count": 3,
 "completed": true,
 "final_output": {
   "tags": [
     "AuthenticationBypass",
     "RequestsPackage",
     "UnauthorizedAccess"
   ],
   "summary": "Critical authentication bypass allows unauthorized access to protected resources and sensitive data via the Requests package."
 },
 "reviewer_feedback": {
   "approved": true,
   "issues": [],
   "latency_ms": 82083.95
 },
 "validation_error": "",
 "token_stats": {
   "turn_count": 2,
   "cumulative_input_tokens": 320,
   "cumulative_output_tokens": 602,
   "cumulative_total_tokens": 922,
   "serialized_history_length": 2
 }
}
C:\Projects\data260-SB44> |
```

Part 4 — Schema Validation and Loop Safety

Schema Validation Experiment

The frozen input was stored in:

reports/hw02/cases/schema_input.json

The input was run 30 times.

Outcome	Count
Valid first attempt	30
Valid after 1 retry	0
Valid after 2 or more retries	0
Hit turn ceiling	0

Completion rate: 100%

Mean latency: 178,399.88 ms

Turn Ceiling Comparison

Each ceiling was tested 20 times using the same frozen input and model settings.

Turn ceiling	Runs	Completed	Completion rate	Mean latency
2	20	0	0%	71,869.60 ms
10	20	20	100%	157,746.71 ms

A ceiling of 2 was too small because the workflow normally requires Supervisor, Planner, and Reviewer turns before it can finish.

A ceiling of 10 completed all runs successfully, so ceiling 10 was selected for deployment.

Adversarial Experiment

The adversarial input was stored in:

`reports/hw02/cases/adversarial_input.json`

It was run five times.

Outcome	Count
Completed	5
Hit turn ceiling	0

Completion rate: 100%

Observed ceiling rate: 0%

Mean latency: 134,891.41 ms

The adversarial input did not reach the turn ceiling. This observed result is reported honestly rather than claiming that the input reliably causes failure.

```
PS C:\Projects\data260-8844> Get-Content ".\reports\hw02\raw\experiment_summary.json"
{
  "schema_validation": {
    "run_count": 30,
    "completion_count": 30,
    "completion_rate": 1.0,
    "mean_latency_ms": 178399.88,
    "classifications": {
      "Valid first attempt": 30
    }
  },
  "ceiling_2": {
    "run_count": 20,
    "completion_count": 0,
    "completion_rate": 0.0,
    "mean_latency_ms": 71869.6,
    "classifications": {
      "Hit turn ceiling": 20
    }
  },
  "ceiling_10": {
    "run_count": 20,
    "completion_count": 20,
    "completion_rate": 1.0,
    "mean_latency_ms": 157746.71,
    "classifications": {
      "Valid first attempt": 20
    }
  },
  "adversarial": {
    "run_count": 5,
    "completion_count": 5,
    "completion_rate": 1.0,
    "mean_latency_ms": 134891.41,
    "classifications": {
      "Valid first attempt": 5
    }
  }
}
PS C:\Projects\data260-8844> |
```

Docker Deployment

The FastAPI application was packaged in a Docker image named:

`data260-8844-hw2`

The container was started with:

```
docker run -d -p 8744:80 --name data260-8844-hw2-container data260-8844-hw2
```

The container started successfully, and the API responded with HTTP 200. The webpage also loaded successfully from the Docker container.

```
PS C:\Projects\data260-8844> docker ps --filter "name=data260-8844-hw2-container"
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
679d003d618d   data260-8844-hw2   "uvicorn web_applica..."   14 hours ago   Up 15 minutes   0.0.0.0:8744->80/tcp, [::]:8744->80/tcp   data260-8844-hw2-container
PS C:\Projects\data260-8844> |
```

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Enter your SJSU email address

Vulnerability Description

Describe the vulnerability and its potential impact

Severity Category

Select severity

☐ I agree to the terms and conditions.

Submit Vulnerability Report

Verification

12 / 14

The verification script is:

`code/verify_hw02.py`

It checks:

- Required HW2 files
- FastAPI API response
- LangGraph smoke-test completion

The final verification produced:

```
{
  "all_passed": true
}
```

```
PS C:\Projects\data260-8844> Get-Content ".\reports\hw02\verification.json"
{
  "homework": "DATA 260 Homework 2",
  "sid4": 8844,
  "port_base": 8744,
  "seed": 8844,
  "verify_seed": 268844,
  "model": "qwen3:8b",
  "timestamp_utc": "2026-09-13T18:59:55.365903+00:00",
  "commit_hash": "e09e8dcfcb2c657a0c20b1019258b2f50b38c9ee",
  "checks": [
    {
      "name": "Required HW2 files",
      "passed": true,
      "details": "All required files exist."
    },
    {
      "name": "FastAPI API response",
      "passed": true,
      "details": "HTTP 200; received 3 reports"
    },
    {
      "name": "LangGraph smoke test",
      "passed": true,
      "details": "Return code 0; GRAPH RESULT found"
    }
  ],
  "all_passed": true
}
PS C:\Projects\data260-8844> |
```

Reproducibility

The following files contain the reproducible HW2 evidence:

- `reports/hw02/RUN_LOG.txt`

- reports/hw02/raw/schema_validation_results.json
- reports/hw02/raw/schema_validation_results.csv
- reports/hw02/raw/ceiling_comparison_results.json
- reports/hw02/raw/ceiling_comparison_results.csv
- reports/hw02/raw/adversarial_results.json
- reports/hw02/raw/experiment_summary.json
- reports/hw02/verification.json

The main commands are:

```
python .\code\run_hw02_experiments.py
python .\code\verify_hw02.py
docker build -f .\code\Dockerfile -t data260-8844-hw2 .
docker run -d -p 8744:80 --name data260-8844-hw2-container data260-8844-hw2
```

Conclusion

This homework extended the HW1 vulnerability-report application into a Dockerized FastAPI application with CRUD operations, search, responsive styling, visible interface states, and a stateful LangGraph agent workflow. The measured experiments showed that a turn ceiling of 10 was reliable for the tested input, while a ceiling of 2 was insufficient.